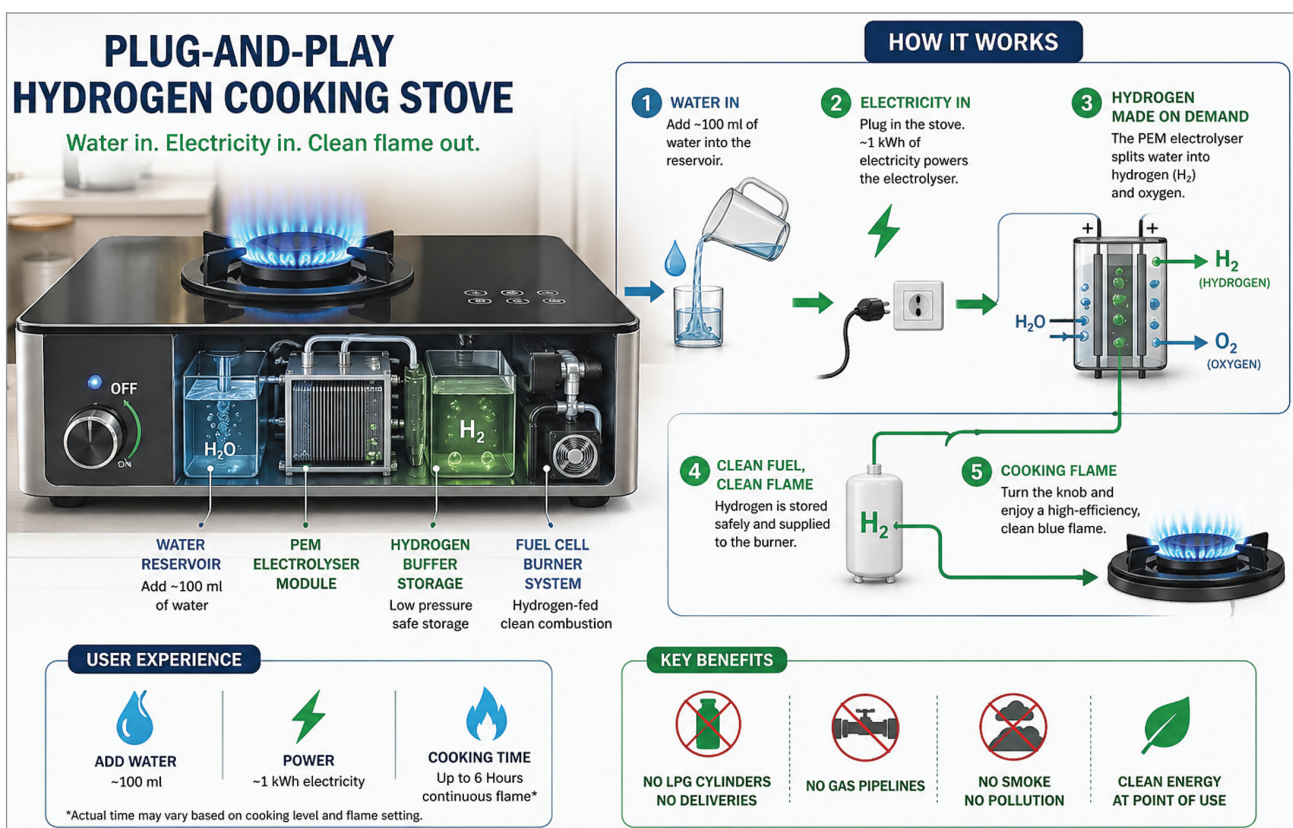


FROM WATER TO FLAME: The Evolution Of Hydrogen Cooking

Imagine cooking with a stove that uses nothing more than water and electricity to generate fuel on demand. With no need for cylinders or pipelines, it challenges conventional ideas about how energy reaches the kitchen. This is not 'burning water,' but a fundamentally different approach to fuel generation. Could it eventually rival LPG and induction cooking in India?



AKANKSHA SONDHI
GAUR

According to the International Energy Agency, almost 2.3 billion people continue to rely on environmentally damaging fuels. As countries move towards sustainable energy sources, there is growing interest in hydrogen as an energy carrier. Hydrogen cookers are now emerging whose primary function is to produce hydrogen fuel from water and electricity, eliminating the need for LPG cylinders or hydrogen pipelines.

In collaboration with India's oil and natural gas industry, Greenvize, a cleantech startup, has developed a portable hydrogen

stove for domestic and commercial applications. At the heart of the device is a proton exchange membrane (PEM) electrolyser integrated into a stove-like appliance.

Water is not the fuel source; it is the feed-stock used to produce hydrogen. Electricity is used to split water molecules and generate hydrogen, which is then combusted. This process introduces energy losses that make the overall system less efficient than LPG and induction cooking. However, a major advantage is that hydrogen is produced continuously on demand, eliminating the